

AP-E4: Tangent-Valued Dirac Operators on $\mathbb{CP}^1$ Chirality, Complete Partner Spectrum, Spectral Gap, and the Canonical-Spin^c versus Ordinary-Spin Gate

Route F research note

15 July 2026

Abstract

This note independently constructs and solves the AP-E4 operator problem. Linearizing a $\mathbb{CP}^1$ sigma-model field proves that its physical horizontal variation is tangent-valued. In affine charts the coefficient transforms by $\delta(1/w) = -w^{-2}\delta w$, hence $T^{1,0}\mathbb{CP}^1 \simeq \mathcal{O}(2)$ with Chern number two. This is a geometric theorem about a bosonic fluctuation; it does not yet produce a fermion.

The minimal mathematical three-mode candidate is the canonical Spin^c Dolbeault operator

$$D_T^c = \sqrt{2}(\bar{\partial}_T + \bar{\partial}_T^\dagger), \quad W^+ = \Omega^{0,0}(T), \quad W^- = \Omega^{0,1}(T).$$

Hodge theory gives $\text{Ker } D_T^{c,+} = H^0(\mathcal{O}(2)) \simeq \mathbb{C}^3$, $\text{Ker } D_T^{c,-} = H^1(\mathcal{O}(2)) = 0$, and index three. In the AP-E1 round sphere convention $R = 1/2$, its complete spectrum is

$$\lambda_{n,\pm} = \pm \frac{\sqrt{n(n+3)}}{R}, \quad \text{mult}(\lambda_{n,+}) = \text{mult}(\lambda_{n,-}) = 2n+3, \quad n \geq 1.$$

Thus the first massive level is ± 4 , fivefold for each sign. Only the kernel lacks opposite chirality; every massive mode has a partner.

A decisive negative control is also proved. The unique ordinary spin structure has $S^+ = \mathcal{O}(-1)$. Twisting an ordinary spin Dirac operator only by the actual tangent line $T = \mathcal{O}(2)$ leaves the Dolbeault line $\mathcal{O}(1)$ and therefore gives two zero modes, not three. The canonical Spin^c choice with $\mathcal{O}(2)$ is equivalent to ordinary spin twist $\mathcal{O}(3)$: its extra half-canonical unit is physical input, not notation.

The companion verifier passes all exact and numerical spectral checks. AP-E4 is therefore mathematically closed at the canonical-operator level, but the physical origin remains open: a bosonic tangent fluctuation does not derive a target-space fermion, a product compactification or moduli-space SQM has not been specified, higher-dimensional anomalies are not cancelled, and the three automorphism vector fields are not yet chiral Route-E families. Physics promotion remains false.

Contents

1	Question, conventions, and exact status	2
2	First-principles derivation of the tangent bundle	2
2.1	Horizontal projection of a projective doublet	2
2.2	Two-chart proof that $T^{1,0}\mathbb{CP}^1 = \mathcal{O}(2)$	3
2.3	The bosonic Hessian is not a Dirac operator	3
3	The canonical Spin^c tangent operator	4
3.1	Operator, chirality, and global bundle	4
3.2	Local ordinary-spin representative and the hidden shift	4

4	Index, chirality, and exact zero modes	5
5	Complete spectrum and spectral gap	5
5.1	Monopole harmonics and the Weitzenböck shift	5
5.2	The tangent result at $q = 2, R = 1/2$	6
5.3	Ordinary spin twisted by the tangent line	6
6	Partner spectrum and possible four-dimensional chirality	6
6.1	What “no opposite chirality” actually means	6
6.2	A conditional product compactification	6
6.3	The automorphism-versus-family gate	7
7	Deterministic analytical and numerical audit	7
8	Claim ledger and next steps	7

1 Question, conventions, and exact status

AP-E4 must keep four statements separate:

- (Q1) Is the linearized physical field tangent-valued?
- (Q2) Which Clifford module and Dirac operator act on it?
- (Q3) What are the chirality, complete spectrum, and first gap?
- (Q4) Under what additional microscopic conditions may the index be read as a four-dimensional family count?

The first three have exact mathematical answers once the operator choice is declared. The fourth is a physical UV problem and is not solved by an index.

We retain the AP-E1 Fubini–Study convention

$$ds^2 = \frac{dw d\bar{w}}{(1 + |w|^2)^2} = R^2(d\theta^2 + \sin^2 \theta d\phi^2), \quad R = \frac{1}{2}. \quad (1)$$

Consequently

$$K_G = \frac{1}{R^2} = 4, \quad \text{Scal} = \frac{2}{R^2} = 8. \quad (2)$$

Every eigenvalue below refers to the canonical geometric normalization $D^c = \sqrt{2}(\bar{\partial} + \bar{\partial}^\dagger)$ and to Eq. (1).

2 First-principles derivation of the tangent bundle

2.1 Horizontal projection of a projective doublet

Let a sigma-model field be represented by

$$z(x) \in \mathbb{C}^2, \quad z^\dagger z = 1, \quad z \sim e^{i\alpha(x)} z. \quad (3)$$

An allowed variation obeys $\text{Re}(z^\dagger \delta z) = 0$. Its vertical gauge component is proportional to iz , while the gauge-horizontal physical variation is

$$\eta = Q_z \delta z, \quad Q_z = 1 - z z^\dagger, \quad z^\dagger \eta = 0. \quad (4)$$

Under $z \mapsto e^{i\alpha}z$, one has $\eta \mapsto e^{i\alpha}\eta$. Intrinsically,

$$T_{[z]}\mathbb{CP}^1 \simeq \text{Hom}(\mathbb{C}z, \mathbb{C}^2/\mathbb{C}z), \quad (5)$$

and η represents exactly this homomorphism. Thus a fluctuation about $\varphi_0 : X \rightarrow \mathbb{CP}^1$ lies in

$$\eta \in \Gamma(X, \varphi_0^* T\mathbb{CP}^1). \quad (6)$$

Notice the pullback: this does not yet make η a spinor living on target $\mathbb{CP}^1$.

2.2 Two-chart proof that $T^{1,0}\mathbb{CP}^1 = \mathcal{O}(2)$

On U_N , set

$$z_N(w) = \frac{(1, w)^T}{\sqrt{1 + |w|^2}}. \quad (7)$$

A direct differentiation and projection gives

$$Q_N \delta z_N = \frac{(-\bar{w}, 1)^T}{(1 + |w|^2)^{3/2}} \delta w. \quad (8)$$

On U_S , use $v = 1/w$ and

$$z_S(v) = \frac{(v, 1)^T}{\sqrt{1 + |v|^2}}, \quad Q_S \delta z_S = \frac{(1, -\bar{v})^T}{(1 + |v|^2)^{3/2}} \delta v. \quad (9)$$

On the overlap,

$$\delta v = \frac{dv}{dw} \delta w = -w^{-2} \delta w. \quad (10)$$

The holomorphic tangent coefficient therefore has degree two. Hence

$$\boxed{T^{1,0}\mathbb{CP}^1 \simeq \mathcal{O}(2)}. \quad (11)$$

For the positive AP-E1 orientation, a compatible Chern connection is

$$a_N^{(2)} = (1 - \cos \theta) d\phi, \quad a_S^{(2)} = -(1 + \cos \theta) d\phi, \quad (12)$$

$$a_N^{(2)} - a_S^{(2)} = 2d\phi, \quad F_T = \sin \theta d\theta \wedge d\phi. \quad (13)$$

Thus

$$\frac{1}{2\pi} \int_{S^2} F_T = 2. \quad (14)$$

The transition proof fixes the algebraic bundle; the flux fixes its oriented connection without relying on ambiguous names for ket versus dual lines.

2.3 The bosonic Hessian is not a Dirac operator

For

$$S_B = \frac{f^2}{2} \int_X |d\varphi|^2, \quad (15)$$

the covariant exponential-map expansion gives

$$S_B^{(2)} = \frac{f^2}{2} \int_X [\langle D_\mu \eta, D^\mu \eta \rangle - \langle R(\eta, \partial_\mu \varphi_0) \partial^\mu \varphi_0, \eta \rangle]. \quad (16)$$

The Jacobi operator is second order,

$$J\eta = -D_\mu D^\mu \eta - R(\eta, \partial_\mu \varphi_0) \partial^\mu \varphi_0 \quad (17)$$

in the displayed Euclidean sign convention. For constant φ_0 , the pullback bundle is trivial on X , the curvature term vanishes, and one gets only $-\square_X$. If φ_0 has degree d on a two-cycle, then $c_1(\varphi_0^* T) = 2d$, not unconditionally two.

Fail-closed gate 2.1 (Statistics and base manifold). The field η is commuting and carries no Clifford index. A fermion requires a separate spin or Spin^c module. In an ordinary supersymmetric sigma model, $\psi \in \Gamma(S_X \otimes \varphi^* T)$ is acted on by a Dirac operator on the base X , not automatically by a Dirac operator whose coordinate is target $\mathbb{CP}^1$. A target spectrum requires an explicit moduli-space quantum mechanics or a product compactification.

3 The canonical Spin^c tangent operator

3.1 Operator, chirality, and global bundle

On a Kähler curve, the canonical Spin^c module is [1, 2]

$$W^+ = \Omega^{0,0}, \quad W^- = \Omega^{0,1}, \quad \Gamma|_{W^\pm} = \pm 1. \quad (18)$$

Twist it by $E = T^{1,0}\mathbb{CP}^1 = \mathcal{O}(2)$. The selected AP-E4 operator is

$$\boxed{D_T^c = \sqrt{2}(\bar{\partial}_T + \bar{\partial}_T^\dagger) : \Omega^{0,0}(T) \oplus \Omega^{0,1}(T) \longrightarrow \Omega^{0,0}(T) \oplus \Omega^{0,1}(T).} \quad (19)$$

It is self-adjoint and odd,

$$\{D_T^c, \Gamma\} = 0, \quad (D_T^c)^2 = 2\Delta_{\bar{\partial}_T}. \quad (20)$$

The canonical Spin^c determinant before twisting is $K^{-1} = \mathcal{O}(2)$. Twisting the module by $E = \mathcal{O}(q)$ changes the effective determinant to

$$\det W_E = K^{-1} \otimes E^2 = \mathcal{O}(2q + 2). \quad (21)$$

For $q = 2$, this is $\mathcal{O}(6)$. This topological datum must eventually be part of the microscopic theory.

3.2 Local ordinary-spin representative and the hidden shift

The unique spin structure on S^2 has

$$S^+ = K^{1/2} = \mathcal{O}(-1), \quad S^- = K^{-1/2} = \mathcal{O}(1). \quad (22)$$

An ordinary spin Dirac operator twisted by $\mathcal{O}(p)$ may be written on the north patch as

$$D_p^{\text{spin}} = -\frac{i}{R} \left[\sigma_1 \left(\partial_\theta + \frac{1}{2} \cot \theta \right) + \frac{\sigma_2}{\sin \theta} \left(\partial_\phi - i a_\phi^{(p)} \right) \right], \quad (23)$$

where

$$a_N^{(p)} = \frac{p}{2}(1 - \cos \theta)d\phi, \quad a_S^{(p)} = -\frac{p}{2}(1 + \cos \theta)d\phi. \quad (24)$$

Its positive component is the Dolbeault line

$$S^+ \otimes \mathcal{O}(p) = \mathcal{O}(p - 1). \quad (25)$$

Consequently

$$D_{\mathcal{O}(q)}^c \simeq D_{\mathcal{O}(q+1)}^{\text{spin}}. \quad (26)$$

The canonical tangent choice $q = 2$ is equivalent to ordinary twist $p = 3$, not $p = 2$. The unit shift is the half-canonical factor $K^{-1/2} = \mathcal{O}(1)$.

Fail-closed gate 3.1 (Ordinary tangent negative control). If a microscopic fermion is an ordinary spinor twisted only by the actual tangent line $T = \mathcal{O}(2)$, then $p = 2$ and its positive Dolbeault line is $\mathcal{O}(1)$. It has two zero modes. Calling this operator “ Spin^c on T ” does not change the count. The physical problem is to derive the canonical Spin^c module or the extra $\mathcal{O}(1)$, not to rename it.

4 Index, chirality, and exact zero modes

Hodge theory gives [2]

$$\text{Ker } D_T^{c,+} \simeq H^0(\mathbb{CP}^1, \mathcal{O}(2)), \quad \text{Ker } D_T^{c,-} \simeq H^1(\mathbb{CP}^1, \mathcal{O}(2)). \quad (27)$$

For $\mathcal{O}(q)$ on $\mathbb{CP}^1$,

$$h^0(\mathcal{O}(q)) = \max\{q+1, 0\}, \quad h^1(\mathcal{O}(q)) = h^0(\mathcal{O}(-q-2)). \quad (28)$$

The second identity is Serre duality. At $q = 2$,

$$h^0(\mathcal{O}(2)) = 3, \quad h^1(\mathcal{O}(2)) = h^0(\mathcal{O}(-4)) = 0. \quad (29)$$

Equivalently, by the index theorem [3], with $x = c_1(\mathcal{O}(1))$ and $\int_{\mathbb{CP}^1} x = 1$,

$$\begin{aligned} \text{ind } D_T^c &= \int_{\mathbb{CP}^1} \text{ch}(\mathcal{O}(2)) \text{Td}(T\mathbb{CP}^1) \\ &= \int_{\mathbb{CP}^1} (1+2x)(1+x) = 3. \end{aligned} \quad (30)$$

Thus

$$\boxed{\dim \text{Ker } D_T^{c,+} = 3, \quad \dim \text{Ker } D_T^{c,-} = 0, \quad \text{ind } D_T^c = 3.} \quad (31)$$

The basis is the familiar $\{z_0^2, z_0 z_1, z_1^2\}$, and $H^0(T\mathbb{CP}^1) \simeq \mathfrak{sl}_2(\mathbb{C})$ as a complex representation.

Remark 4.1 (Orientation reversal is shifted). For the canonical Spin^c operator, the index of $\mathcal{O}(q)$ is $q+1$. Three negative-chirality zero modes require $q = -4$, not $q = -2$. This asymmetry is another manifestation of the built-in canonical determinant. An ordinary spin monopole spectrum has the more familiar sign-symmetric charge count after its half-canonical shift is included explicitly.

5 Complete spectrum and spectral gap

5.1 Monopole harmonics and the Weitzenböck shift

For a canonical Dolbeault twist $E = \mathcal{O}(q)$, $q \geq 0$, the relevant $SU(2)$ multiplet at excitation number $n \geq 0$ has

$$j = \frac{q}{2} + n, \quad \dim V_j = 2j+1 = q+2n+1. \quad (32)$$

The connection Laplacian has eigenvalue

$$\nabla^* \nabla = \frac{1}{R^2} \left[j(j+1) - \left(\frac{q}{2} \right)^2 \right]. \quad (33)$$

On positive chirality, the Kähler–Weitzenböck identity subtracts the curvature term $q/(2R^2)$:

$$(D_{\mathcal{O}(q)}^c)^2 = 2\bar{\partial}_q^\dagger \bar{\partial}_q = \nabla^* \nabla - \frac{q}{2R^2}. \quad (34)$$

Substitution of Eq. (32) gives the exact identity

$$j(j+1) - \frac{q^2}{4} - \frac{q}{2} = n(n+q+1). \quad (35)$$

At $n = 0$ this vanishes and the multiplicity is $q+1$, reproducing the index result.

For $n \geq 1$, supersymmetry/Hodge decomposition pairs the positive and negative form sectors. Since D is odd and self-adjoint, every positive eigenvalue has a negative partner. The full spectrum is therefore

$$\boxed{\lambda_{n,\pm} = \pm \frac{\sqrt{n(n+q+1)}}{R}, \quad \text{mult}(\lambda_{n,\pm}) = q+2n+1, \quad n \geq 1.} \quad (36)$$

This agrees with the supersymmetric projective-space monopole spectrum after converting operator and radius normalizations [4].

5.2 The tangent result at $q = 2, R = 1/2$

For AP-E4,

$$\lambda_{n,\pm} = \pm 2\sqrt{n(n+3)}, \quad \text{mult}(\lambda_{n,\pm}) = 2n + 3. \quad (37)$$

The first levels are

n	$ \lambda_n $	multiplicity	interpretation
0	0	3 in W^+ , 0 in W^-	chiral kernel; no $\pm\lambda$ split
1	4	5	first massive partner level
2	$2\sqrt{10}$	7	massive partner level
3	$6\sqrt{2} \simeq 8.48528$	9	massive partner level

Thus

$$\boxed{\Delta_D = 4, \quad \Delta_{D^2} = 16.} \quad (38)$$

If the microscopic kinetic operator carries a velocity or normalization v_D , all masses are multiplied by v_D ; the dimensionless geometric gap is the value above.

5.3 Ordinary spin twisted by the tangent line

For ordinary $p = 2$, the effective Dolbeault degree is $q = p - 1 = 1$. Hence

$$\dim \text{Ker } D_{p=2}^{\text{spin},+} = 2, \quad \lambda_{n,\pm}^{\text{spin}} = \pm \frac{\sqrt{n(n+p)}}{R}, \quad \text{mult} = p + 2n. \quad (39)$$

At $R = 1/2$,

$$\Delta_{p=2}^{\text{spin}} = 2\sqrt{3} \simeq 3.464101615, \quad \text{mult}_{n=1} = 4 \quad (40)$$

per sign. This different kernel, gap, and degeneracy make the hidden Spin^c choice experimentally and numerically falsifiable.

6 Partner spectrum and possible four-dimensional chirality

6.1 What “no opposite chirality” actually means

Equation (31) proves that the *kernel* has no negative-chirality state. It does not remove negative chirality from the theory. For every $n \geq 1$, choose normalized chiral components $\psi_n^{(0)} \in W^+$ and $\psi_n^{(1)} \in W^-$ so that

$$D_T^c \psi_n^{(0)} = |\lambda_n| \psi_n^{(1)}, \quad D_T^c \psi_n^{(1)} = |\lambda_n| \psi_n^{(0)}. \quad (41)$$

The operator then produces the normalized eigenstates

$$\psi_{n,+} = \frac{\psi_n^{(0)} + \psi_n^{(1)}}{\sqrt{2}}, \quad \psi_{n,-} = \frac{\psi_n^{(0)} - \psi_n^{(1)}}{\sqrt{2}}, \quad (42)$$

with eigenvalues $\pm|\lambda_n|$. The first massive level contains five such pairs. Any portal with characteristic scale comparable to $4v_D$ will mix these states into the purported three-mode sector.

6.2 A conditional product compactification

Only after declaring a product geometry $M^{3,1} \times \mathbb{CP}^1$ may one write schematically

$$\not{D}_6 = \not{D}_4 \otimes 1 + \gamma_5 \otimes D_T^c, \quad \Gamma_7 = \gamma_5 \otimes \Gamma_2, \quad (43)$$

up to a conventional overall sign. A single six-dimensional Weyl condition then ties internal to four-dimensional chirality: the three internal positive zero modes become three same-chirality

four-dimensional massless modes. Each nonzero internal pair becomes a massive four-dimensional Dirac pair. A six-dimensional *Dirac* field contains both six-dimensional chiralities and generally doubles the four-dimensional zero sector.

Fail-closed gate 6.1 (Higher-dimensional anomalies). A single six-dimensional Weyl fermion is generally gauge- and gravitationally anomalous. A physical compactification must cancel its full eight-form anomaly polynomial, or exhibit a valid Green–Schwarz factorization, and must quantize the Spin^c determinant flux. After reduction, the representation-weighted four-dimensional anomaly is multiplied by the index three. These are spectrum-level consistency conditions, not optional phenomenology [5].

6.3 The automorphism-versus-family gate

The vector space

$$H^0(\mathbb{CP}^1, T\mathbb{CP}^1) \simeq \mathfrak{sl}_2(\mathbb{C}) \quad (44)$$

consists of holomorphic infinitesimal automorphisms of $\mathbb{CP}^1$. Depending on the microscopic construction, these may be gauge redundancies, collective coordinates, or physical matter modes. Dimension three alone cannot decide. One must derive the kinetic norm, gauge quotient, couplings, and anomaly representation of each mode before calling them three families.

7 Deterministic analytical and numerical audit

The companion script `route_f/code/verify_ap_e4_tangent_dirac_spectrum.py` is deliberately independent of a fitted spectrum. It performs:

1. exact line-bundle cohomology and Serre-duality checks for $q = 2$, ordinary $p = 2$, and neighbouring $q = 1, 3$ controls;
2. exact rational verification of Eq. (35) and its multiplicity through ten levels;
3. numerical construction of each finite $SU(2)$ spin- j matrix, diagonalization of the shifted Casimir, and comparison with $n(n + q + 1)$;
4. a truncated chiral block operator with three positive zero bases and five massive levels, testing Hermiticity, $\{D, \Gamma\} = 0$, exact $\pm\lambda$ pairing, kernel chirality, and the gap four;
5. curvature quadrature for $c_1(T) = 2$, horizontal-projection residuals, and radius-rescaling checks;
6. negative controls for nonintegral monopole charge, ordinary-spin misidentification, a chirality-breaking mass, and the false assertion that massive partners are absent.

The spectral block regression checks bookkeeping, while the independent Casimir diagonalization checks the representation-theoretic eigenvalues. Neither is presented as a substitute for a microscopic origin.

8 Claim ledger and next steps

Claim	Status	Boundary
Horizontal projective fluctuation is tangent-valued	proved	as a section of $\varphi_0^* T\mathbb{CP}^1$
$T^{1,0}\mathbb{CP}^1 = \mathcal{O}(2)$, $c_1 = 2$	proved	chart and flux proofs

Claim	Status	Boundary
Canonical Spin^c tangent operator has three positive zero modes	proved	declared operator choice
Complete canonical spectrum and gap four	proved	AP-E1 $R = 1/2$ convention
Every massive level has paired $\pm D_T^c$ eigenvalues and equal $W^\pm$ multiplicities	proved	only the kernel has a chiral imbalance
Ordinary spin twist by T has two zero modes	proved	decisive negative control
Physical fermion is canonical Spin^c and tangent-valued	open	requires moduli SQM or compactification
Six-dimensional and four-dimensional anomalies cancel	open	no complete field content or I_8 factorization
Three zero modes are three Route-E families	open	automorphism, gauge, and portal gates unresolved

Three physically distinct continuations should be retained:

1. **Moduli-space/SQM route.** Derive a soliton moduli space $\mathbb{CP}^1$, its tangent fermions, and $N = 2$ supercharge $Q = \sqrt{2}\bar{\partial}_T$; then compute whether these quantum states carry the required four-dimensional gauge representation.
2. **Product-compactification route.** Specify a six-dimensional anomaly-free theory on $M^{3,1} \times \mathbb{CP}^1$, derive the Spin^c determinant line and $E = T$, and calculate the complete KK/threshold spectrum.
3. **Ordinary-spin control route.** Accept the natural ordinary tangent twist and its two modes unless an additional $\mathcal{O}(1)$ gauge flux is independently derived. This is the conservative fallback.

The fail-closed result is

```

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canonical_spinc_three_zero_modes=true,
partner_spectrum_and_gap_complete=true,
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six_dimensional_anomaly_cancelled=false,
route_e_chiral_family_identification=false,
ap_e4_physics_closed=false,
physics_promotion_allowed=false.

```

(45)

References

- [1] H. Blaine Lawson and Marie-Louise Michelsohn. *Spin Geometry*. Princeton University Press, 1989.
- [2] Raymond O. Wells. *Differential Analysis on Complex Manifolds*. Springer, third edition, 2008.
- [3] Michael F. Atiyah and Isadore M. Singer. The index of elliptic operators: I. *Annals of Mathematics*, 87(3):484–530, 1968.

- [4] Dmitri Bykov and Andrei Smilga. Monopole harmonics on complex projective spaces. *SciPost Physics*, 15:195, 2023.
- [5] Luis Alvarez-Gaumé and Edward Witten. Gravitational anomalies. *Nuclear Physics B*, 234:269–330, 1984.